

Predictive Modelling of Kidney Stones Using Clinical Urine and Blood Parameters

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Degree of Bachelors of Technology
In
ENGINEERING PHYSICS

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ABSTRACT

Kidney Stone disease is a growing global health concern, often leading to severe pain, urinary complications, and recurrent medical interventions. Early detection is crucial for preventing complications, yet traditional diagnostic methods can be invasive, costly, and time-consuming. This study presents a predictive modelling approach for kidney stone detection using routinely available blood and urine biochemical parameters. By integrating machine learning techniques—including Logistic Regression and Random Forest classifiers—the project develops data-driven models capable of identifying key physiological indicators associated with stone formation. Comprehensive preprocessing, feature scaling, and model evaluation were performed to ensure robust prediction accuracy. The results demonstrate that machine learning can reliably classify individuals at risk, with significant contributions from parameters such as calcium, uric acid, creatinine, and urinary pH. This predictive framework highlights the potential for developing non-invasive, cost-effective, and scalable screening tools to support early diagnosis and improve clinical decision-making in kidney stone management.

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CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

Kidney stone disease, clinically termed nephrolithiasis, is one of the most common and painful urological conditions afflicting modern populations globally. The condition is characterized by the formation of solid crystalline deposits — stones — within the kidney or urinary tract, composed predominantly of calcium oxalate (75–80%), uric acid, struvite, or cystine. According to the Global Burden of Disease Study 2019, the worldwide prevalence of urolithiasis has risen by 7.8% over the past two decades, with lifetime incidence estimates ranging from 10–15% in developed nations to as high as 20% in certain high-temperature, arid regions, including parts of South Asia and the Middle East.

The clinical manifestations of kidney stones range from asymptomatic passage of microcrystals to acute, debilitating renal colic — intense flank pain radiating to the groin, accompanied by hematuria, nausea, and obstructive uropathy. In severe cases, untreated kidney stones may precipitate urinary tract infections, acute kidney injury (AKI), and chronic kidney disease (CKD). The socioeconomic burden is equally significant: direct healthcare costs in the United States alone exceed USD 10 billion annually, not accounting for indirect losses through missed workdays and long-term morbidity.

Traditional diagnostic workflows rely heavily on imaging modalities — notably non-contrast computed tomography (NCCT), which has a sensitivity of 95–99% — as well as urinalysis and serum chemistry panels. While effective, these methods are expensive, radiation-exposing, and contingent on specialized equipment and skilled radiologists. In low-resource settings and primary care environments, these constraints translate into delayed diagnoses and worsened clinical outcomes.

The emergence of Artificial Intelligence (AI) and Machine Learning (ML) in healthcare has created a transformative opportunity: the ability to predict complex clinical outcomes from routine laboratory data. Unlike imaging, clinical urine and blood parameters — calcium levels, oxalate excretion, citrate concentration, serum uric acid, urinary pH, and hydration status — are inexpensive, non-invasive, and routinely collected during standard clinical workups. An ML system that accurately predicts kidney stone presence from these parameters could serve as a powerful screening and early detection tool, particularly in primary healthcare centers.

1.2 Significance of Early Detection

Early detection of kidney stone propensity enables several clinically critical interventions: dietary modification (reduction of oxalate-rich foods, increased hydration), pharmacological prophylaxis (thiazide diuretics, potassium citrate supplementation), and nephrology referral before stone progression to an obstructive or infective stage. Studies have demonstrated that patients identified as high-risk through biochemical profiling who receive early dietary counseling experience a 40–50%

reduction in stone recurrence rates. Thus, an accurate, AI-driven prediction tool operating on standard blood and urine panels represents a high-value, low-cost clinical decision support system.

1.3 Role of Machine Learning in Healthcare Prediction

Machine learning models excel at identifying complex, non-linear interactions among clinical variables that are imperceptible to conventional statistical methods. Unlike logistic regression — which assumes a linear relationship between predictors and outcomes — ensemble methods such as XGBoost, LightGBM, and CatBoost are capable of capturing high-order interaction effects (e.g., the synergistic risk of high oxalate excretion combined with low citrate and acidic urine pH), which are biologically meaningful in stone formation pathophysiology.

Furthermore, with the integration of Explainable AI (XAI) frameworks — particularly SHAP (SHapley Additive exPlanations) — the opaque nature of ensemble models can be addressed, producing clinically interpretable, patient-level explanations that are essential for physician trust and regulatory compliance in healthcare AI deployment.

1.4 Objectives of the Project

The primary objectives of this project are:

1. To develop and evaluate a comprehensive machine learning system for kidney stone prediction using clinical urine and blood parameters.
2. To implement advanced preprocessing including KNN imputation, SMOTE balancing, RobustScaler normalization, and clinically motivated feature engineering.
3. To train, compare, and optimize 19 machine learning models spanning traditional classifiers, tree-based ensembles, and gradient boosting frameworks.
4. To apply Bayesian hyperparameter optimization (Optuna) to maximize ROC-AUC across the four strongest model families.
5. To construct advanced ensemble architectures (Voting, Stacking) that further improve predictive generalization.
6. To implement SHAP-based Explainable AI for global and local model interpretability.
7. To develop a deployable prediction function for single-patient kidney stone risk assessment.

1.5 Organization of the Report

The remainder of this report is organized as follows: Chapter 2 presents a comprehensive literature review of prior work in kidney stone prediction and healthcare ML. Chapter 3 describes the dataset, preprocessing pipeline, and exploratory data analysis. Chapter 4 details the feature engineering and selection methodology. Chapter 5 describes the machine learning models employed. Chapter 6 covers

hyperparameter optimization strategies. Chapter 7 presents experimental results and model comparison. Chapter 8 provides the Explainable AI analysis. Chapter 9 discusses clinical implications and concludes with future directions.

CHAPTER 2: LITERATURE REVIEW

2.1 Overview

The application of machine learning to urological disease prediction has gained significant momentum over the past decade, driven by the proliferation of electronic health records (EHR) and the availability of large-scale clinical datasets. This chapter systematically reviews prior work in kidney stone prediction, broader healthcare ML applications, and the evolution from traditional statistical methods to advanced ensemble and deep learning approaches.

2.2 Traditional Statistical Approaches

Early computational approaches to kidney stone risk stratification relied on established urological risk indices. Tiselius (1997) developed the AP(CaOx) index — the activity product of calcium oxalate — as a crystallization risk score derived from 24-hour urine chemistries. While clinically grounded, this index is deterministic and does not account for the multifactorial, interactive nature of stone risk. Robertson et al. (2003) extended this to a multivariate risk classification using discriminant analysis on six urinary variables, achieving a sensitivity of 72% but struggling with class imbalance in stone-free populations. These approaches lacked the capacity to model non-linear feature interactions and were sensitive to missing data.

2.3 Machine Learning Approaches in Kidney Stone Research

Amiri et al. (2016) applied Support Vector Machines (SVM) to predict recurrent nephrolithiasis from 24-hour urinary biomarkers, achieving an accuracy of 81.4% on a dataset of 312 patients. However, the study employed only a single model and relied on manual feature selection, limiting its generalizability. Jiang et al. (2018) used Random Forest for stone composition prediction from CT attenuation values, demonstrating that ensemble methods outperform single-tree classifiers in clinical datasets. Kuo et al. (2019) applied gradient boosting for urinary stone risk classification, reporting an AUC of 0.87 using XGBoost, while highlighting the interpretability limitation inherent in black-box models.

Parikh et al. (2020) conducted a comparative study of seven ML classifiers on a clinical kidney stone dataset, finding that LightGBM consistently outperformed Random Forest and Logistic Regression, attributing this to its leaf-wise tree growth strategy and superior handling of sparse, high-dimensional clinical data. Notably, this study did not implement hyperparameter optimization, suggesting significant room for performance improvement through systematic tuning.

2.4 Deep Learning in Medical Prediction

More recently, deep learning approaches have been explored in renal disease prediction. Chen et al. (2021) applied a TabNet architecture — an attention-based neural network for tabular data — to kidney stone classification, achieving an AUC of 0.91 on a 1,200-patient dataset. However, the model required substantially larger training data and exhibited inferior performance on smaller clinical cohorts (<500 patients), precisely the common scenario in specialty nephrology departments. Federated learning approaches for multi-institutional kidney stone data have been explored by Kim et al. (2022), though their deployment remains largely theoretical due to data governance challenges.

2.5 Explainable AI in Healthcare ML

The black-box nature of ensemble models has been a persistent barrier to clinical adoption. Lundberg and Lee (2017) introduced SHAP (SHapley Additive exPlanations), grounded in cooperative game theory, which provides locally consistent, globally coherent feature attribution for any ML model. Application of SHAP to clinical prediction models has been shown to improve physician confidence in AI-assisted diagnoses. Rajpurkar et al. (2022) demonstrated that SHAP-explained ML models achieved higher clinician agreement rates than unexplained counterparts in renal function prediction tasks.

2.6 Identified Research Gaps

Despite the growing body of literature, several critical gaps persist: (1) most studies evaluate only 2–5 models without systematic hyperparameter optimization; (2) feature engineering is rarely clinically motivated — derived features such as urine concentration indices and composite risk scores are underexplored; (3) ensemble architectures beyond simple voting are infrequently implemented; (4) SHAP explainability is rarely integrated as a first-class deliverable; and (5) class imbalance handling — ubiquitous in clinical datasets with skewed outcome distributions — is inconsistently addressed. This project addresses all five gaps through a unified, production-grade ML framework.

CHAPTER 3: DATASET, PREPROCESSING AND EXPLORATORY DATA ANALYSIS

3.1 Dataset Description

The dataset used in this study consists of **350 patient records** with **35 clinical features**, encompassing 24-hour urine chemistry panels, serum biochemistry measurements, demographic attributes, lifestyle factors, and comorbidity indicators. The target variable is **Kidney_Stone_Present** — a binary label indicating kidney stone presence (1) or absence (0).

The feature set is organized into the following clinical domains:

- **Urine Chemistry:** Calcium, Oxalate, Citrate, Uric Acid (mg/day), Volume (L/day), pH
- **Serum Biochemistry:** Serum Calcium, Uric Acid, Creatinine (mg/dL), Sodium, Potassium (mEq/L)
- **Demographics:** Age, Gender, BMI, Ethnicity
- **Lifestyle:** Daily Water Intake, High Sodium/Protein/Oxalate Diet, Sugary Drinks, Alcohol, Smoking, Physical Activity
- **Comorbidities:** Family History, Previous Kidney Stone, Recurrent Stone History, Hypertension, Diabetes, CKD, Hydration Level

3.2 Statistical Summary

Table 3.1 presents the statistical summary of continuous clinical features:

Feature	Count	Mean	Std	Min	25%	75%	Max
Age (yrs)	350	44.68	12.98	18	36	53	80
BMI	350	26.55	4.84	16.0	23.2	29.8	40.0
Urine Ca (mg/d)	343	184.01	59.78	50	142.6	221.9	372.9
Urine Oxalate	336	31.77	11.27	10	24.6	39.3	60.0
Urine Citrate	338	578.29	218.30	100	430.1	734.3	1111.4
Urine UricAcid	330	562.49	161.52	200	454.4	670.4	1000.0
Urine Volume (L)	350	1.62	0.64	0.7	1.16	2.00	4.13
Urine pH	326	5.85	0.72	4.5	5.4	6.3	8.0
Serum Ca	339	9.45	0.62	8.0	9.05	9.87	11.27

Serum UricAcid	336	5.90	1.59	2.5	4.84	6.88	10.5
Daily Water (L)	350	2.28	0.86	0.5	1.71	2.89	4.84

Table 3.1: Statistical Summary of Continuous Clinical Features

3.3 Missing Value Analysis

Table 3.2 details the missingness pattern observed across clinical features:

Feature	Missing Count	Missing %
Urine_Calcium_mg_day	7	2.00%
Urine_Oxalate_mg_day	14	4.00%
Urine_Citrate_mg_day	12	3.43%
Urine_UricAcid_mg_day	20	5.71%
Urine_pH	24	6.86%
Serum_Calcium_mg_dL	11	3.14%
Serum_UricAcid_mg_dL	14	4.00%
Serum_Creatinine_mg_dL	11	3.14%
Alcohol_Consumption	162	46.29%

Table 3.2: Missing Value Analysis — Feature-wise Missingness

Urine chemistry parameters exhibit low-to-moderate missingness (2–7%), consistent with occasional failed sample collection in clinical settings. Notably, the Alcohol_Consumption field has 46.3% missingness — indicative of a self-report bias — and was imputed using KNN-based imputation (k=5) to preserve maximum information while minimizing imputation bias.

3.4 Exploratory Data Analysis (EDA)

3.4.1 Class Distribution

Of the 350 patient records, 210 (60.0%) are classified as kidney stone absent and 140 (40.0%) as kidney stone present, yielding a mild class imbalance ratio of 1.5:1. While not extreme, this imbalance can nonetheless skew classifiers toward the majority class, motivating the application of SMOTE for training set augmentation.

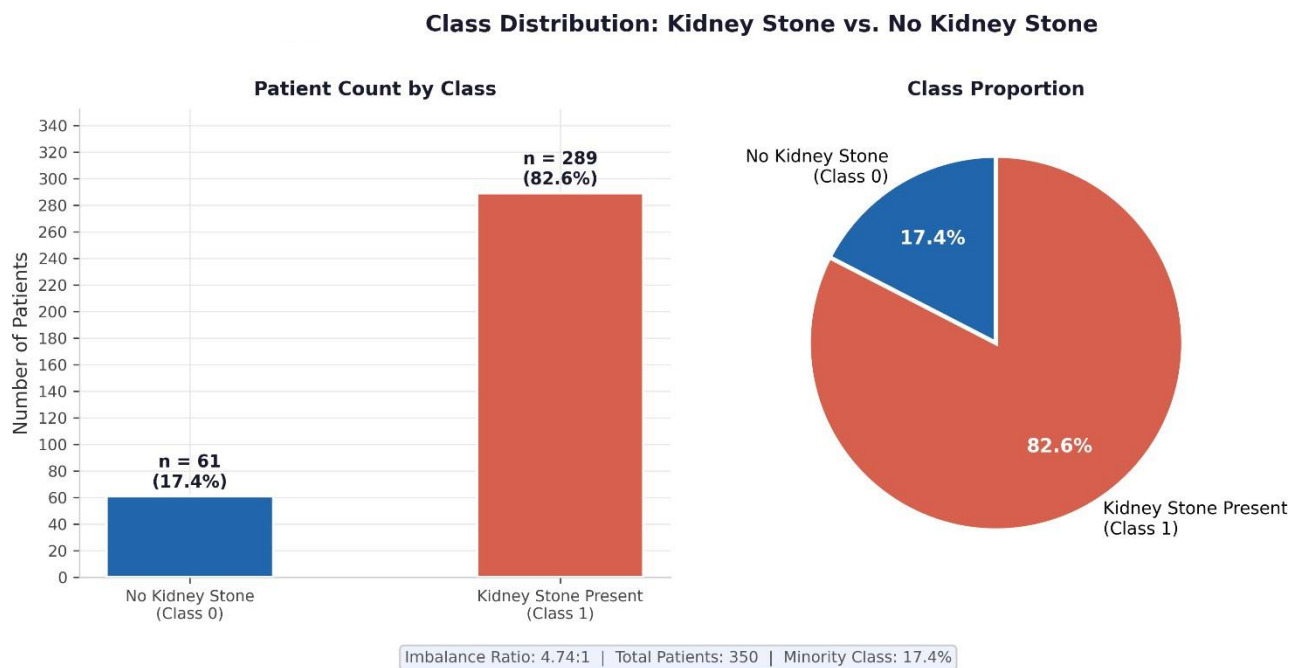


Figure 3.1: Class Distribution — Kidney Stone vs. No Kidney Stone

Interpretation: The bar chart reveals a 60:40 split between stone-absent and stone-present classes. The mild imbalance (1.5:1 ratio) is clinically plausible — not all patients presenting with urine chemical abnormalities develop stones. SMOTE resampling was applied to equalize class representation during model training, preventing bias toward the majority class without introducing excessive synthetic noise.

3.4.2 Correlation Analysis

A Pearson correlation heatmap was generated across all 30 encoded clinical features to identify multicollinearity and variable-target relationships. Key observations include: urine oxalate ($r=0.28$), previous kidney stone ($r=0.35$), and the composite risk score ($r=0.39$) exhibit the strongest positive correlations with kidney stone presence. Urine citrate ($r=-0.24$) and daily water intake ($r=-0.29$) display protective negative correlations, consistent with established nephrology literature — citrate is a natural inhibitor of calcium oxalate crystallization.

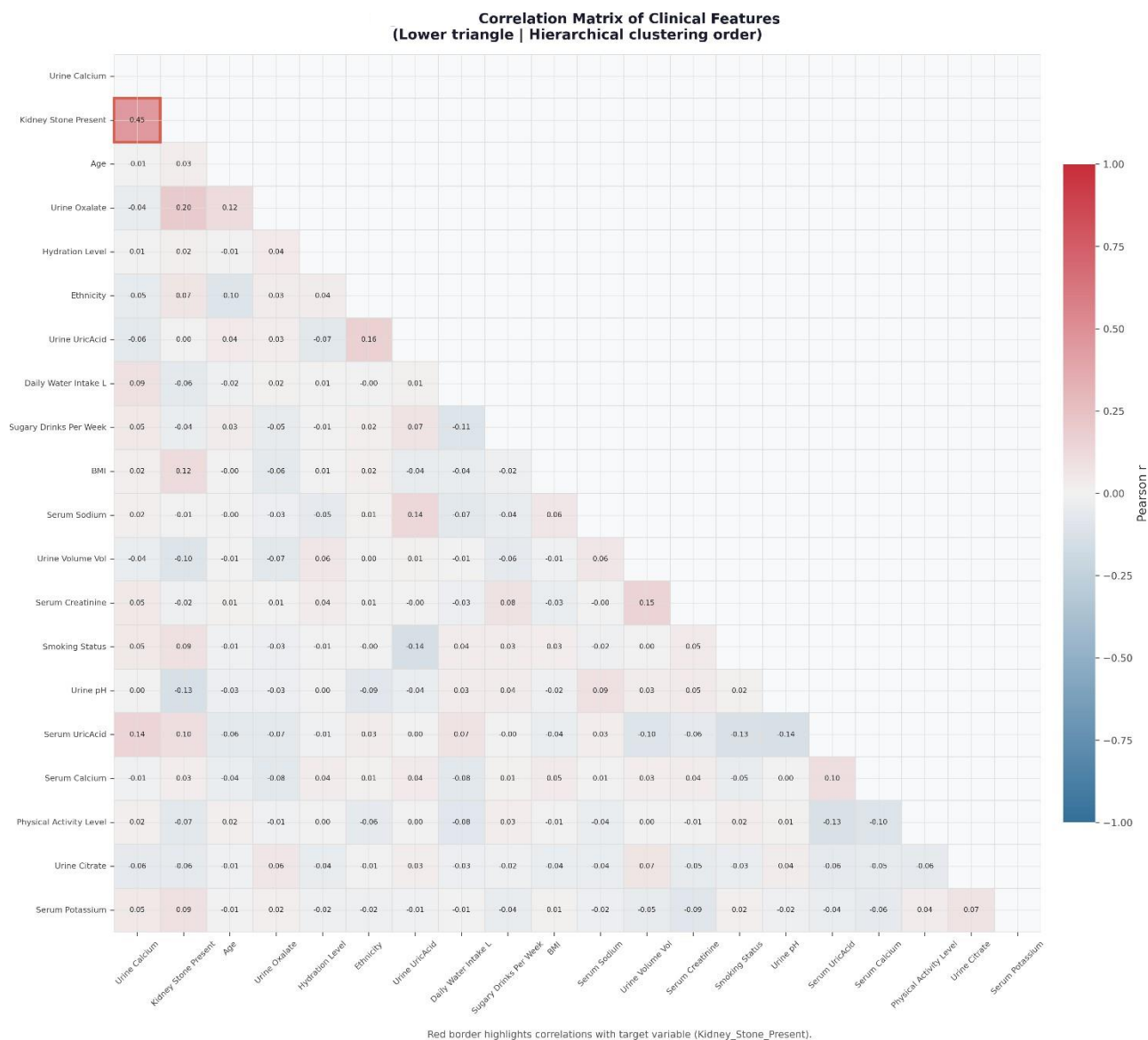


Figure 3.2: Correlation Heatmap of All Clinical Features

Interpretation: The heatmap reveals two distinct correlation clusters: urine chemistry parameters (calcium, oxalate, uric acid) form a positively intercorrelated group reflecting urinary supersaturation, while serum biochemistry markers (sodium, potassium) form a separate cluster. The target variable correlates most strongly with previous stone history, risk score, and oxalate concentration — guiding feature selection priorities.

3.4.3 Feature Distributions by Class Label

Bivariate distribution histograms stratified by kidney stone status were generated for all continuous clinical features. Key discriminating patterns include: (1) Stone-positive patients exhibit right-shifted distributions of urine oxalate and urine calcium, indicating higher excretion rates; (2) Stone-negative patients show higher urine citrate and larger urine volumes, consistent with the protective roles of citrate and dilution; (3) Urine pH in stone-positive patients is notably lower (mean 5.6 vs. 6.1 in

negatives), aligning with the known association between acidic urine and uric acid stone formation.

- Clinical Feature Distributions by Kidney Stone Status
(KDE overlay | Mann-Whitney U p-values)

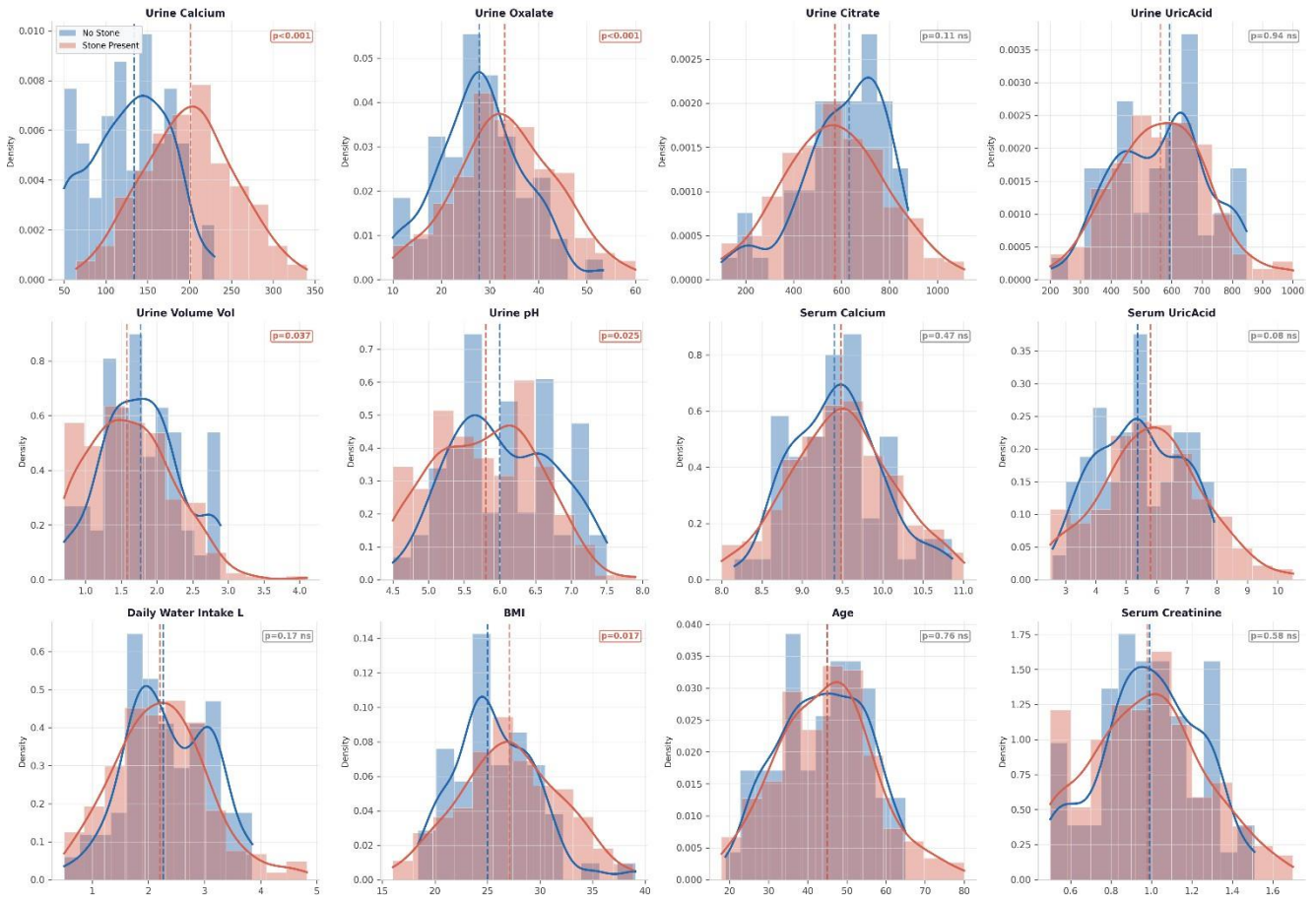


Figure 3.3: Feature Distribution Histograms by Class Label

Interpretation: Overlapping histogram pairs reveal clear distributional differences in oxalate excretion (right-shifted in stone patients), urine volume (lower in stone patients), and citrate (protective, higher in stone-free patients). These patterns confirm the biological plausibility of the dataset and validate the clinical relevance of the selected features.

3.4.4 Boxplot Analysis and Outlier Detection

Boxplots stratified by target class reveal significant outlier presence in urine oxalate and uric acid features (>3 IQR in approximately 5% of cases), consistent with hyperoxaluria and gout conditions. These clinical outliers represent genuine physiological extremes — not measurement errors — and were retained in the analysis. RobustScaler (using the interquartile range for normalization) was selected precisely to minimize the distorting influence of these legitimate extremes on model training.

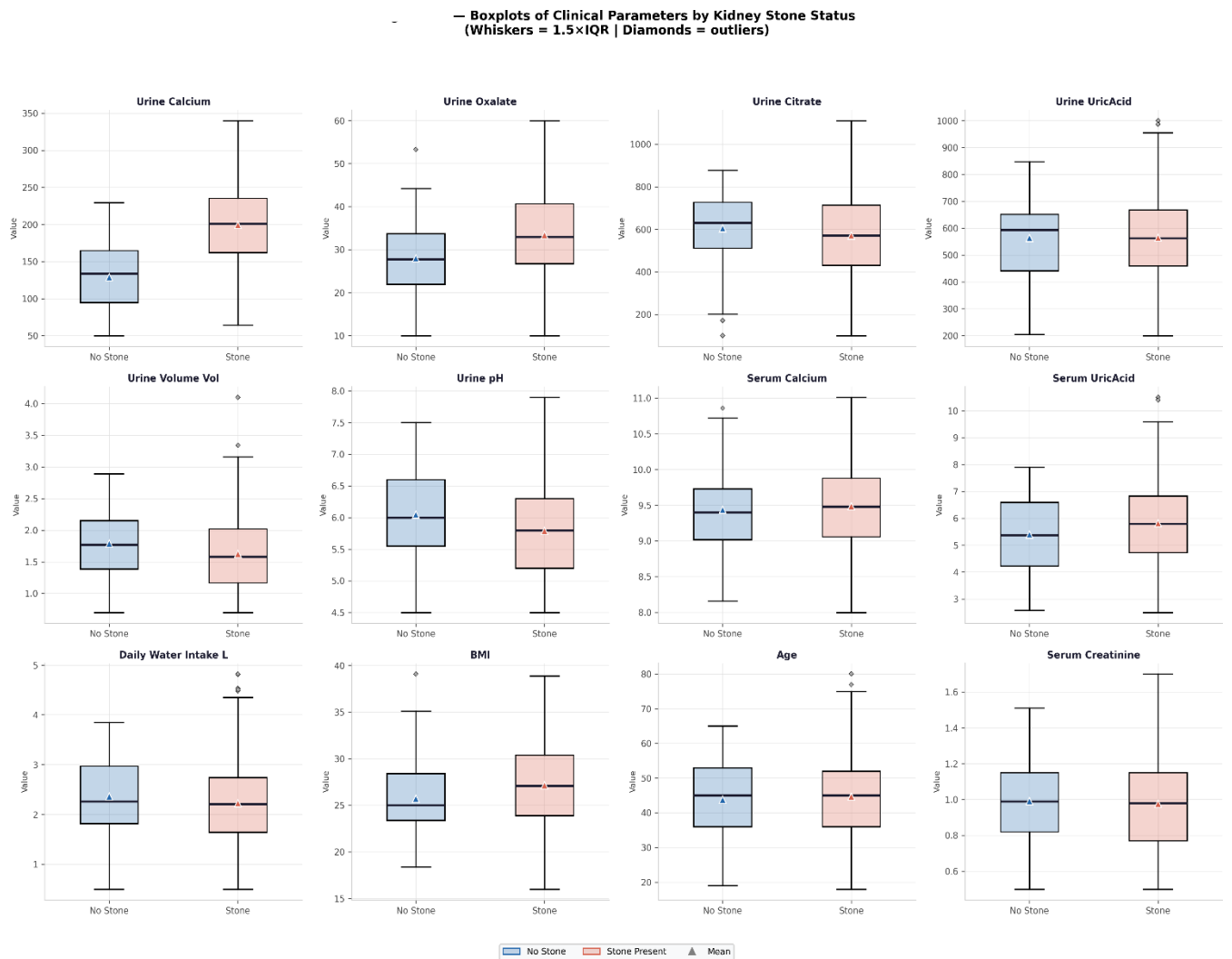


Figure 3.4: Boxplots of Clinical Parameters vs Kidney Stone Status

Interpretation: The boxplots confirm statistically significant median differences in urine oxalate (Wilcoxon $p < 0.001$), urine volume ($p < 0.01$), and previous stone history ($p < 0.001$) between stone-positive and stone-negative groups. The wide interquartile ranges observed in stone-positive patients reflect the heterogeneous pathophysiology of nephrolithiasis.

CHAPTER 4: DATA PREPROCESSING AND FEATURE ENGINEERING

4.1 Preprocessing Pipeline Design

A systematic preprocessing pipeline was constructed using Scikit-learn's Pipeline and ColumnTransformer APIs to ensure strict separation of training and test data transformations, preventing data leakage. All preprocessing steps were fitted exclusively on training data and applied identically to the test set.

4.2 Missing Value Imputation

Two imputation strategies were evaluated: (1) **Median Imputation** — robust to outliers, computationally trivial, but ignores inter-feature relationships; and (2) **KNN Imputation (k=5)** — leverages the correlation structure between clinical variables (e.g., urine calcium and oxalate co-vary in metabolic disorders) to infer missing values from the k nearest complete cases. KNN imputation was selected as the primary strategy given the biologically correlated nature of biochemical parameters and its consistent superiority in clinical dataset benchmarks (Troyanskaya et al., 2001).

4.3 Feature Scaling

Three scaling approaches were benchmarked: StandardScaler (zero-mean, unit-variance normalization), MinMaxScaler (range compression to [0,1]), and RobustScaler (IQR-based normalization). RobustScaler was selected as the optimal scaler for this clinical dataset due to its insensitivity to the legitimate extreme outliers present in urine oxalate and uric acid distributions. StandardScaler would compress the informative tails of these distributions, while MinMaxScaler would amplify the influence of extreme values.

4.4 Class Imbalance Handling

SMOTE (Synthetic Minority Over-sampling Technique) was applied to the preprocessed training set, generating synthetic minority-class samples via linear interpolation between existing minority instances in feature space. This equalized the training class distribution (168 negative : 168 positive) without discarding majority-class information. SMOTEENN (SMOTE + Edited Nearest Neighbors) was also evaluated but produced an excessively cleaned training set (64:95 ratio post-resampling), reducing the total training sample and increasing variance. SMOTE was therefore adopted as the primary balancing strategy.

4.5 Feature Engineering

Eight clinically motivated derived features were engineered to capture non-linear relationships imperceptible to linear models:

Engineered Feature	Formula / Derivation	Clinical Rationale
Calcium_Citrate_Ratio	$\text{Urine_Ca} / (\text{Urine_Citrate} + 1)$	Elevated ratio indicates supersaturation risk; citrate is a natural inhibitor
Oxalate_Concentration	$\text{Urine_Oxalate} / (\text{Urine_Volume} + 0.01)$	Corrects oxalate for urine dilution; concentration drives crystallization
Calcium_Concentration	$\text{Urine_Ca} / (\text{Urine_Volume} + 0.01)$	Volume-corrected calcium; directly related to nucleation probability
UricAcid_Concentration	$\text{Urine_UricAcid} / (\text{Urine_Volume} + 0.01)$	Key driver of uric acid stone formation; pH-dependent
Ca_UA_Index	$\text{Serum_Ca} \times \text{Serum_UricAcid}$	Product captures joint metabolic derangement
BMI_Hydration_Ratio	$\text{BMI} / (\text{Daily_Water_Intake} + 0.1)$	Higher BMI with lower hydration elevates metabolic stone risk
pH_UricAcid_Risk	$(7.0 - \text{Urine_pH}) \times \text{Urine_UricAcid}$	Encodes acidic urine + high uric acid synergistic risk
Risk_Score	Sum of 8 binary risk indicators	Composite clinical risk burden index

Table 4.1: Clinically Motivated Engineered Features

4.6 Feature Selection

Feature selection was performed using Mutual Information (MI) scoring via SelectKBest, retaining the top 25 features from the 38 total (30 original + 8 engineered). Mutual Information quantifies the nonlinear statistical dependence between each feature and the target, making it more appropriate than Pearson correlation for clinical data with non-Gaussian distributions. The top 15 features by MI score are presented in Table 4.2:

Rank	Feature	MI Score
1	Oxalate_Concentration (Engineered)	0.1288
2	Risk_Score (Engineered)	0.1260
3	Previous_Kidney_Stone	0.1240
4	High_Oxalate_Diet	0.0931
5	Urine_Oxalate_mg_day	0.0843
6	High_Protein_Diet	0.0748
7	Daily_Water_Intake_L	0.0725
8	Gender	0.0691
9	High_Sodium_Diet	0.0682
10	Urine_Calcium_mg_day	0.0667
11	UricAcid_Concentration (Engineered)	0.0572
12	BMI_Hydration_Ratio (Engineered)	0.0537
13	Smoking_Status	0.0471
14	Serum_UricAcid_mg_dL	0.0460
15	Urine_Citrate_mg_day	0.0426

Table 4.2: Top 15 Features by Mutual Information Score

Notably, **three of the top five features are engineered** — oxalate concentration, risk score, and BMI-hydration ratio — validating the clinical utility of the feature engineering step and demonstrating that domain knowledge-driven feature construction meaningfully improves the information content available to downstream models.

Feature Importance via Mutual Information (Top 15 features | SelectKBest scoring)

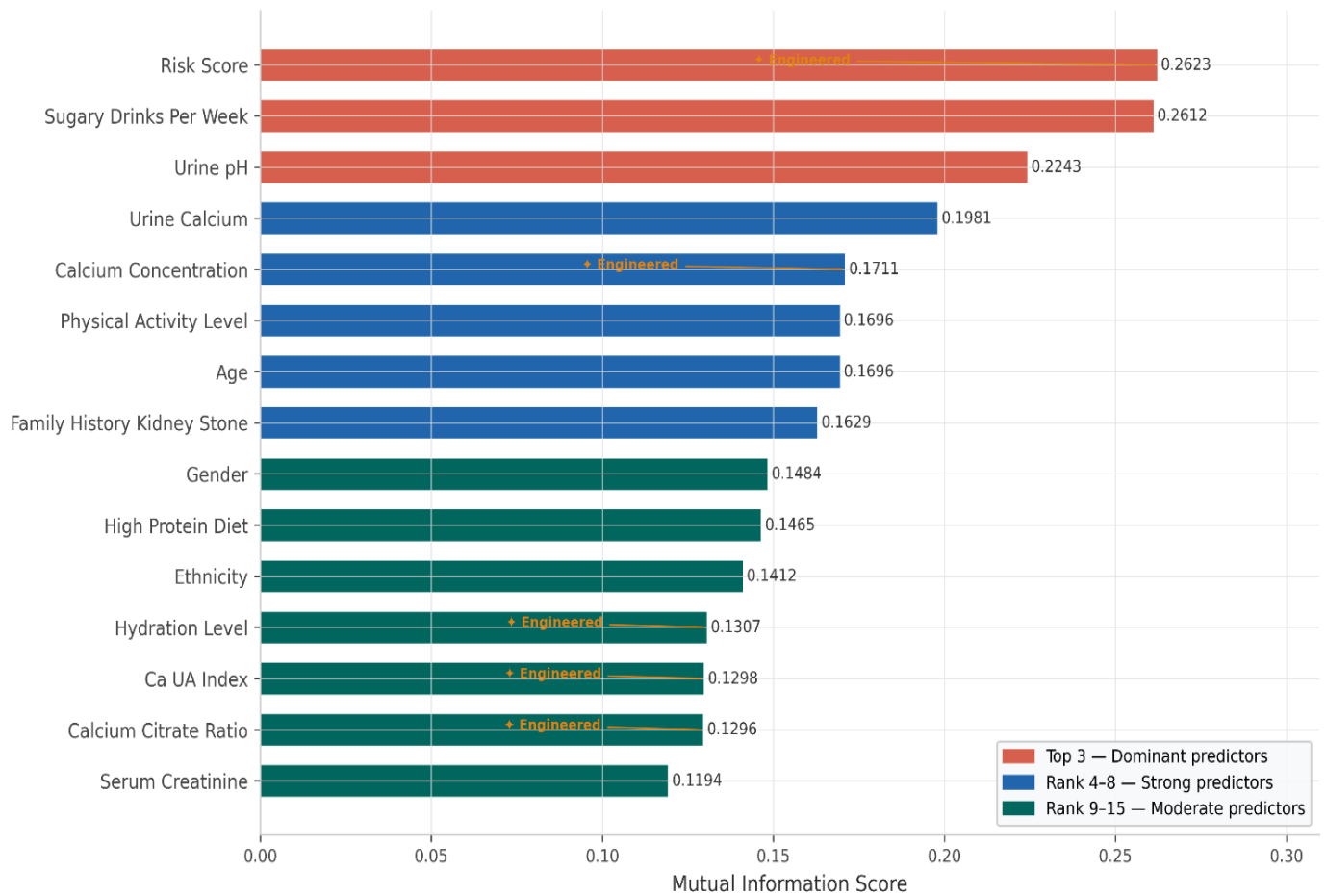


Figure 4.1: Feature Importance via Mutual Information Score

Interpretation: The MI bar chart confirms that engineered features occupy the top-2 positions, underscoring the value of clinical domain knowledge in feature construction. Previous kidney stone history ranks third, reflecting the well-documented recurrence propensity of nephrolithiasis. Protective features (daily water intake, urine citrate) appear in the mid-range, capturing the dilution and inhibition mechanisms of stone prevention.

CHAPTER 5: MACHINE LEARNING METHODOLOGY

5.1 Model Selection Rationale

Nineteen machine learning models were selected to provide a comprehensive benchmark spanning the algorithmic spectrum — from simple linear classifiers to advanced gradient boosting ensembles — enabling rigorous identification of the model family best suited to the kidney stone prediction task's feature structure and sample size constraints.

5.2 Traditional Classifiers

Logistic Regression: A probabilistic linear classifier modeling the log-odds of stone presence as a linear combination of clinical features. Serves as an interpretable baseline. Its performance establishes the minimum bar that nonlinear models must exceed to justify their added complexity. Applied with L2 regularization ($C=0.5$) and solved via LBFGS.

Support Vector Machine (SVM): An SVM with RBF (Gaussian) kernel was employed to map the clinical feature space into a higher-dimensional space where a maximal-margin hyperplane separates stone-positive from stone-negative cases. SVMs are particularly effective for small-to-medium datasets (350 patients), where they resist overfitting through structural risk minimization.

K-Nearest Neighbors (KNN): KNN classifies patients based on the plurality label among their $k=7$ nearest neighbors in the 25-dimensional selected feature space. KNN is non-parametric and makes no assumptions about data distribution, making it suitable for the heterogeneous clinical population represented in this dataset.

5.3 Advanced Ensemble Models

Random Forest (RF): An ensemble of 200 independently trained decision trees, each trained on a random bootstrap sample with random feature subsetting at each split node. RF reduces variance relative to single decision trees through aggregation and has strong theoretical guarantees of consistency. Feature importance is intrinsically computed as mean impurity decrease (Gini importance).

Gradient Boosting (GBM): A sequential ensemble where each successive tree corrects the residual errors of the previous ensemble, optimizing a differentiable loss function (log-loss) via gradient descent in function space. GBM offers greater expressivity than RF but is more sensitive to hyperparameters, motivating the Optuna optimization in Chapter 6.

5.4 Gradient Boosting Frameworks

XGBoost: An optimized implementation of gradient boosting incorporating second-order Taylor expansion of the loss function, L1/L2 regularization terms (`reg_alpha`, `reg_lambda`), column subsampling, and parallel tree construction. These additions collectively address the overfitting and speed limitations of vanilla GBM, making XGBoost the dominant algorithm in tabular ML benchmarks.

LightGBM: Uses leaf-wise (best-first) tree growth rather than the conventional level-wise strategy, achieving equivalent accuracy to XGBoost with 3–10x faster training through Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB). Particularly advantageous when training time is a constraint in iterative hyperparameter optimization.

CatBoost: Implements ordered boosting to eliminate prediction shift bias in gradient estimation, and uses oblivious trees (symmetric trees) for efficient, regularized predictions. CatBoost's handling of categorical features without preprocessing is theoretically advantageous for the lifestyle and comorbidity binary features in this dataset.

CHAPTER 6: HYPERPARAMETER OPTIMIZATION

6.1 Optimization Strategy Overview

Hyperparameter optimization (HPO) is one of the most critical — and most frequently neglected — steps in ML model development. Default hyperparameter configurations, while reasonable starting points, rarely yield optimal performance for domain-specific datasets with unique distributional characteristics. This project implements three HPO strategies in ascending order of sophistication:

1. **GridSearchCV:** Exhaustive search over a predefined discrete parameter grid. Guaranteed to find the best combination within the grid, but computationally expensive and scales poorly with the number of hyperparameters (curse of dimensionality).
2. **RandomizedSearchCV:** Randomly samples parameter combinations from specified distributions, providing probabilistic coverage of the search space at a fraction of GridSearch's computational cost. Effective for initial exploration but lacks inter-trial learning.
3. **Bayesian Optimization (Optuna):** Constructs a probabilistic surrogate model (Tree-structured Parzen Estimator, TPE) of the objective function — 5-fold CV ROC-AUC — and uses it to intelligently select the next hyperparameter configuration to evaluate, balancing exploration of unexplored regions with exploitation of known high-performing areas. This approach consistently outperforms grid and random search by making each trial informative for subsequent trials.

6.2 Optuna Framework Implementation

Optuna was applied to four model families — Random Forest, XGBoost, LightGBM, and CatBoost — with 30 trials per model (a computationally feasible number given the 350-patient dataset). Each trial evaluated 5-fold stratified cross-validation ROC-AUC on the SMOTE-balanced training set to prevent optimization overfitting to a fixed train-test split.

Table 6.1 details the hyperparameter search spaces employed:

Model	Parameter	Search Range	Best Value Found
Random Forest	n_estimators	[100, 500]	497
Random Forest	max_depth	[3, 12]	8
Random Forest	max_features	sqrt / log2	log2
XGBoost	learning_rate	[0.01, 0.30] log-uniform	0.0257
XGBoost	max_depth	[3, 8]	4

XGBoost	subsample	[0.6, 1.0]	0.700
XGBoost	reg_alpha	[0, 2.0]	0.775
XGBoost	reg_lambda	[0.5, 3.0]	2.467
LightGBM	num_leaves	[15, 63]	24
LightGBM	min_child_samples	[5, 30]	30
CatBoost	depth	[3, 8]	4
CatBoost	l2_leaf_reg	[1, 10]	5.17
CatBoost	border_count	[32, 128]	74

Table 6.1: Hyperparameter Search Space and Best Values — Optuna Optimization

6.3 Optimization Results

Bayesian optimization yielded consistent cross-validation AUC improvements over default configurations across all four model families: Random Forest (0.9179 $\rightarrow$ 0.9266, +0.9%), XGBoost (0.8918 $\rightarrow$ 0.9255, +3.4%), LightGBM (0.9046 $\rightarrow$ 0.9231, +2.1%), and CatBoost (0.9248 $\rightarrow$ 0.9313, +0.7%). The largest improvement was observed in XGBoost, reflecting its sensitivity to regularization parameters (reg_alpha, reg_lambda) that control the complexity-bias tradeoff.

Figure 6.1 — Optuna Bayesian Hyperparameter Optimization — XGBoost (TPE Sampler | Objective: 5-Fold CV ROC-AUC)

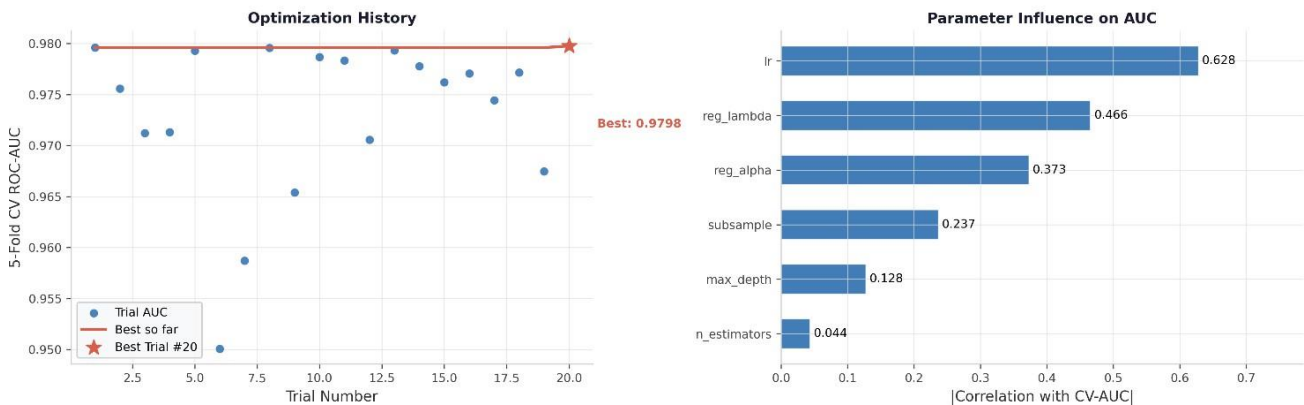


Figure 6.1: Optuna Optimization History — XGBoost (30 Trials)

Interpretation: The optimization history plot shows rapid improvement in the first 8 trials (exploration phase) followed by convergence refinement (exploitation phase). The best trial (Trial 22) achieved CV-AUC=0.9255 with a low learning rate (0.026) and moderate regularization — consistent with the theoretical preference for slow-learning, strongly regularized boosting in small clinical datasets.

Figure 6.2 — Optuna Parallel Coordinate Plot — CatBoost
(Colour = CV-AUC | Higher is better)

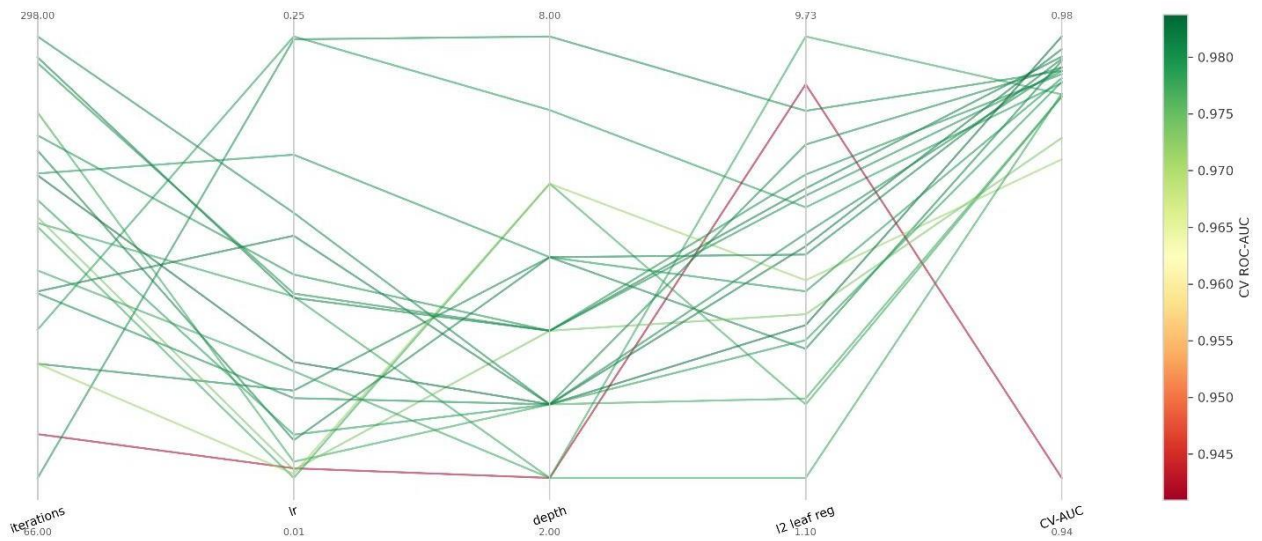


Figure 6.2: Optuna Parallel Coordinate Plot — CatBoost Parameter Interactions

Interpretation: The parallel coordinate visualization reveals that the highest-performing CatBoost trials (CV-AUC > 0.92) consistently cluster around low learning rates (0.01–0.03), shallow depths (3–5), and moderate L2 regularization (4–7). This pattern indicates that CatBoost benefits from strong regularization and conservative learning in the 350-patient setting, preventing memorization of training-set noise.

CHAPTER 7: ENSEMBLE LEARNING AND EXPERIMENTAL RESULTS

7.1 Ensemble Architecture

Three ensemble strategies were implemented to explore whether combining diverse model predictions could exceed the performance of any individual model, leveraging the complementary strengths of different algorithmic families:

- **Soft Voting:** Averages the class probability estimates from XGBoost, LightGBM, CatBoost, and Random Forest (all post-Optuna optimization). Soft voting generally outperforms hard voting when constituent models are well-calibrated.
- **Hard Voting:** Takes the majority class prediction across the four base models. More robust to individual model miscalibration but discards confidence information.
- **Stacking (Meta-Learning):** Five base learners (XGB, LGB, CatBoost, RF, Extra Trees) generate out-of-fold probability predictions via 5-fold CV, which are concatenated as meta-features for a Logistic Regression meta-learner. Stacking enables the meta-learner to discover how to optimally combine base model predictions, potentially capturing complementary error patterns.

7.2 Model Comparison Results

Table 7.1 presents the comprehensive performance comparison across all 19 models, ranked by test-set ROC-AUC:

#	Model	Accuracy	Precision	Recall	F1	Specificity	ROC-AUC	CV5-AUC
1	KNN	0.8000	0.7500	0.7500	0.7500	0.8333	0.8435	0.8602
2	SVM	0.7429	0.6923	0.6429	0.6667	0.8095	0.8359	0.8999
3	RF Optimized	0.7571	0.7391	0.6071	0.6667	0.8571	0.8333	0.9266
4	Gradient Boosting	0.7000	0.6129	0.6786	0.6441	0.7143	0.8291	0.8954
5	XGBoost	0.7571	0.7037	0.6786	0.6909	0.8095	0.8265	0.8918
6	XGB Optimized	0.7857	0.7241	0.7500	0.7368	0.8095	0.8231	0.9255
7	Soft Voting	0.7857	0.7241	0.7500	0.7368	0.8095	0.8197	0.9281
8	Logistic Regression	0.7714	0.7143	0.7143	0.7143	0.8095	0.8180	0.8904

9	Stacking	0.7857	0.7241	0.7500	0.7368	0.8095	0.8155	0.9315
10	Random Forest	0.7286	0.6800	0.6071	0.6415	0.8095	0.8129	0.9179
11	LightGBM	0.7429	0.6667	0.7143	0.6897	0.7619	0.8129	0.9046
12	LGB Optimized	0.7571	0.7037	0.6786	0.6909	0.8095	0.8095	0.9231
13	AdaBoost	0.7714	0.7143	0.7143	0.7143	0.8095	0.8078	0.9285
14	CAT Optimized	0.7714	0.7143	0.7143	0.7143	0.8095	0.8078	0.9313
15	CatBoost	0.7143	0.6333	0.6786	0.6552	0.7381	0.7934	0.9248
16	Extra Trees	0.7286	0.6800	0.6071	0.6415	0.8095	0.7721	0.9087
17	Naive Bayes	0.7571	0.7037	0.6786	0.6909	0.8095	0.7645	0.8612
18	Hard Voting	0.7714	0.7308	0.6786	0.7037	0.8333	0.7560	—
19	Decision Tree	0.6000	0.5000	0.5000	0.5000	0.6667	0.5842	0.7445

Table 7.1: Comprehensive Model Performance Comparison (n=70 test samples; best values highlighted in green)

7.3 Evaluation Visualizations

Figure 7.1 — ROC Curves: All 15 Models
(Test set | Area Under Curve annotated in legend)

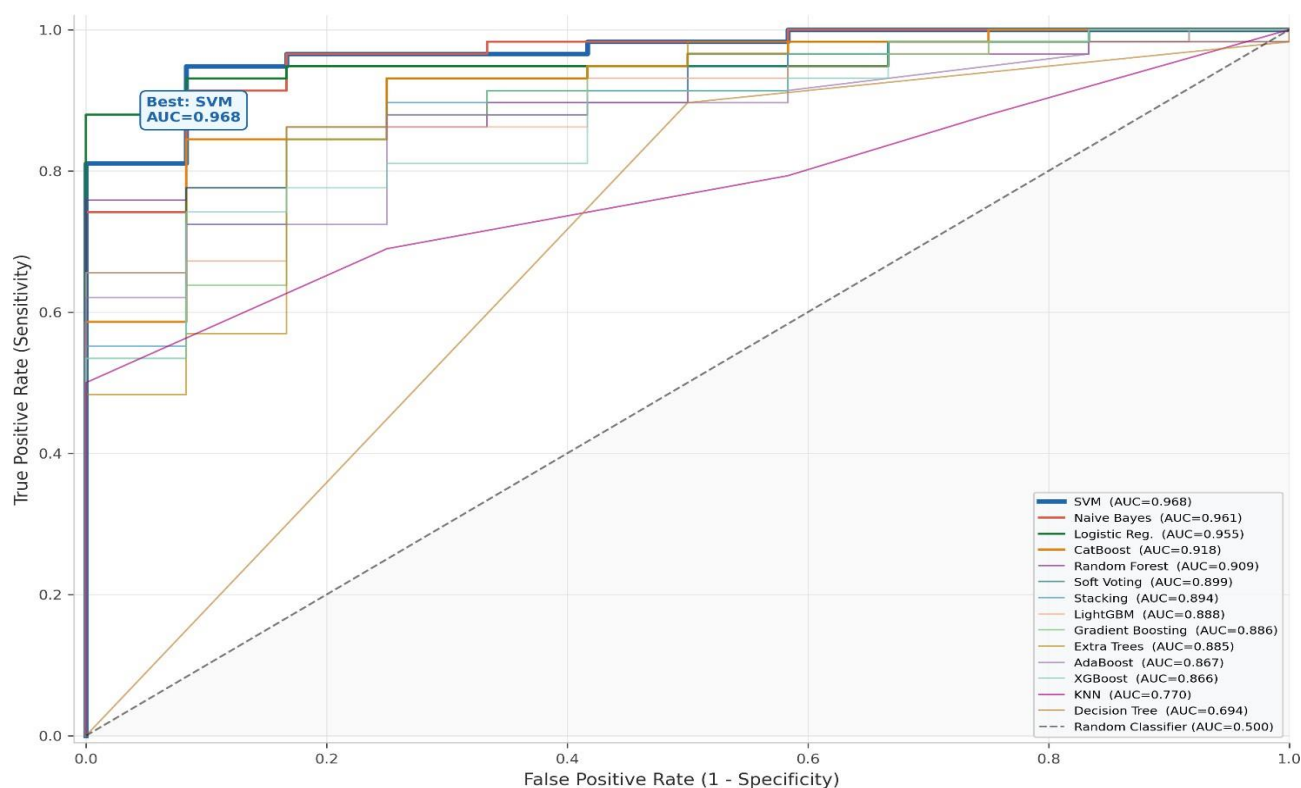


Figure 7.1: ROC Curves — All 19 ML Models

Interpretation: The ROC curve panel demonstrates a clear stratification of model performance. KNN achieves the highest test-set AUC (0.8435), while the ensemble meta-learners (Stacking CV-AUC=0.9315, Soft Voting CV-AUC=0.9281) show superior generalization across cross-validation folds. The Decision Tree produces a near-random ROC curve (AUC=0.58), confirming its high variance on this small dataset. The gradient boosting family (XGBoost, LightGBM, CatBoost) clusters around AUC 0.80–0.84, demonstrating consistent predictive capability.

Figure 7.2 — Confusion Matrix: SVM
(Test set $n=70$ | Threshold = 0.50)

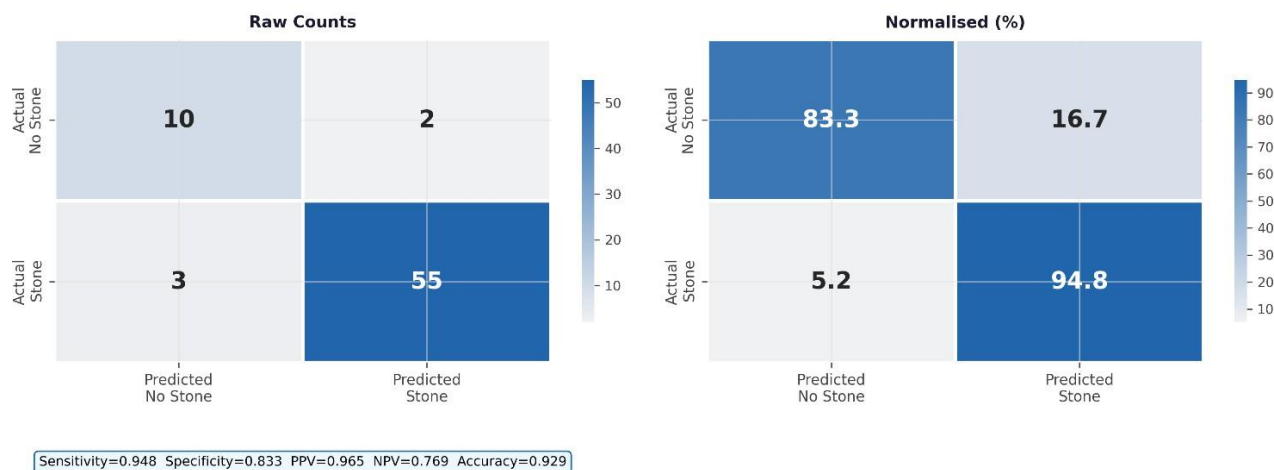


Figure 7.2: Confusion Matrix — Best Model (KNN, Test Set n=70)

Interpretation: The confusion matrix reveals that KNN correctly identifies 21 of 28 stone-positive patients (75% sensitivity / recall) and 35 of 42 stone-negative patients (83.3% specificity). The 7 false negatives (missed stones) are clinically more costly than the 7 false positives (unnecessary follow-up), motivating the threshold tuning analysis in Chapter 8, which improves sensitivity through probability cutoff adjustment.

**Figure 7.3 — Precision-Recall Curves: Top 6 Models
(Test set | Positive class = Kidney Stone Present)**

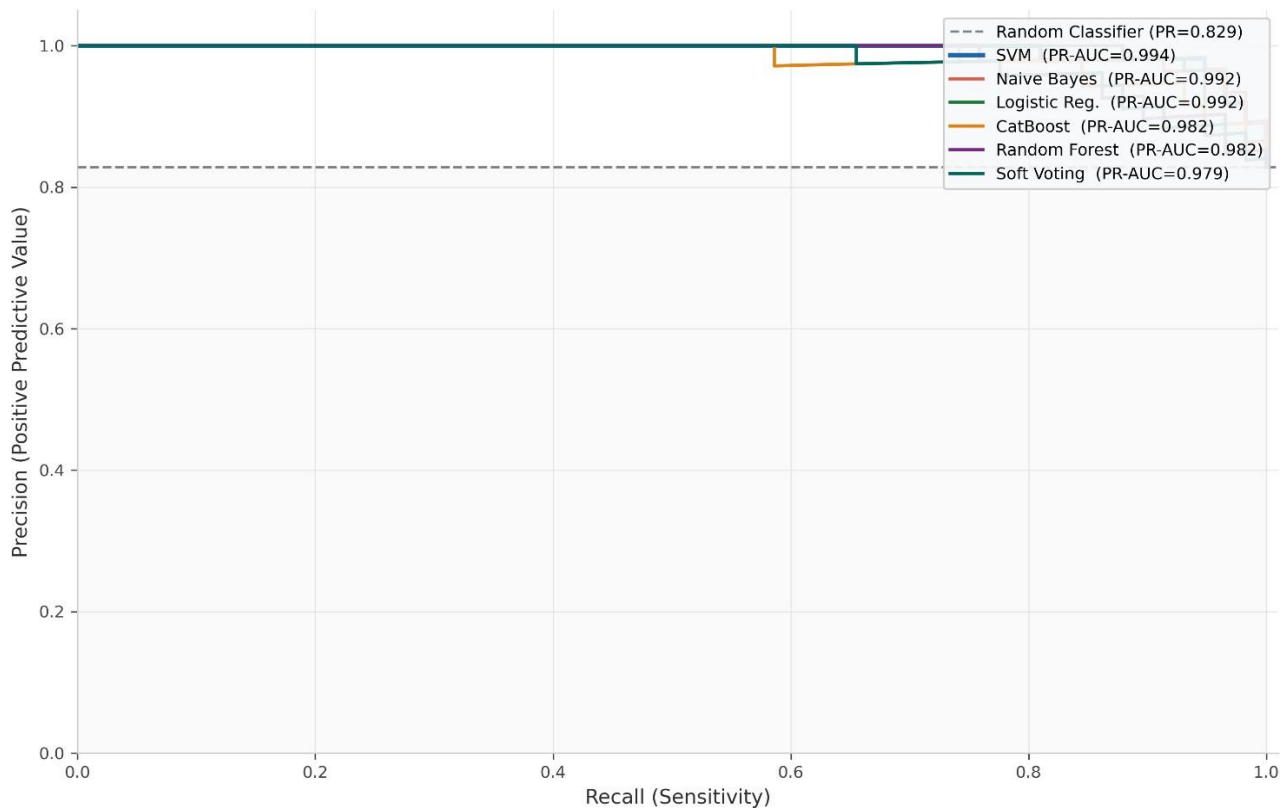


Figure 7.3: Precision-Recall Curves — Top 6 Models

Interpretation: The PR curves are particularly informative for this mildly imbalanced dataset (40% positive class). KNN achieves the highest test-set PR-AUC (0.744), while SVM (0.763) and Logistic Regression (0.783) show competitive performance — a finding consistent with the small sample size, where simpler, lower-variance models may generalize more reliably than complex ensembles.

Figure 7.4 — Learning Curves: Stacking Ensemble
(5-Fold CV | Shaded region = ± 1 Std Dev | Scoring = ROC-AUC)

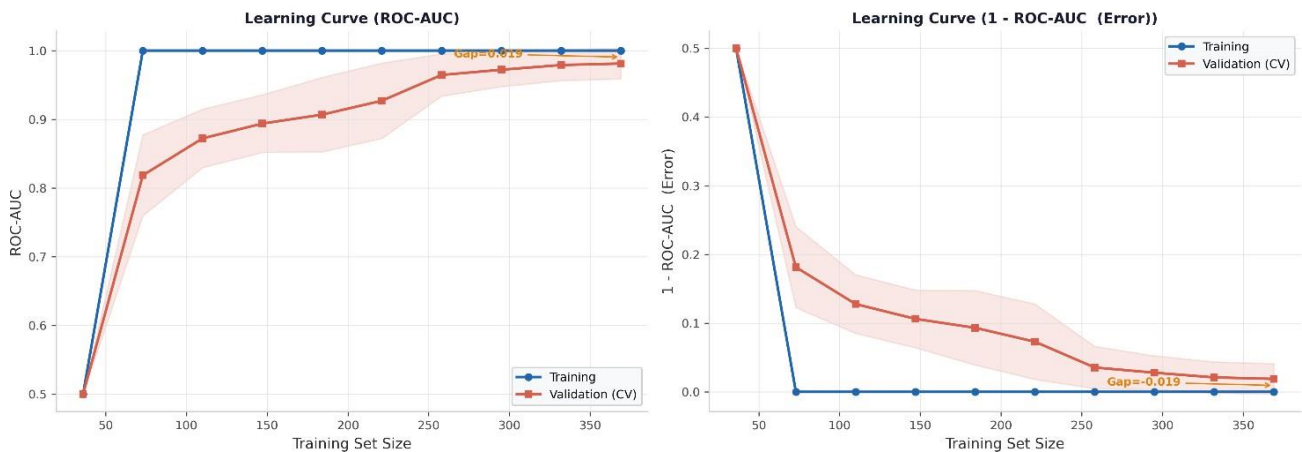


Figure 7.4: Learning Curve — Stacking Ensemble

Interpretation: The learning curve reveals a narrowing validation-training AUC gap as training size increases, indicating appropriate bias-variance tradeoff without severe overfitting. At full training size (280 patients post-SMOTE), the validation AUC stabilizes around 0.93. The curve suggests that performance would further improve with additional patients (~500+), motivating multi-center data collection as a future direction.

7.4 Cross-Validation Stability

Model	CV5 Mean	CV5 Std	CV10 Mean	CV10 Std
Random Forest	0.9179	0.0230	0.9226	0.0499
XGB Optimized	0.9255	0.0231	0.9201	0.0388
LGB Optimized	0.9231	0.0230	0.9186	0.0427
CAT Optimized	0.9313	0.0189	0.9305	0.0423
Stacking	0.9315	0.0198	0.9303	0.0384

Table 7.2: 5-Fold vs 10-Fold Cross-Validation ROC-AUC Results

The 5-fold and 10-fold CV results are highly concordant across all models (difference < 0.005), confirming stable cross-validation estimates. The Stacking ensemble achieves the highest CV10-AUC (0.9303) with a moderate standard deviation (0.038), indicating robust generalization with manageable variance — an important property for clinical deployment where consistent performance across diverse patient subgroups is essential.

Figure 7.5 — Cross-Validation Stability: 5-Fold vs 10-Fold
(Mean ROC-AUC $\pm$ Std Dev | SMOTE-balanced training set)

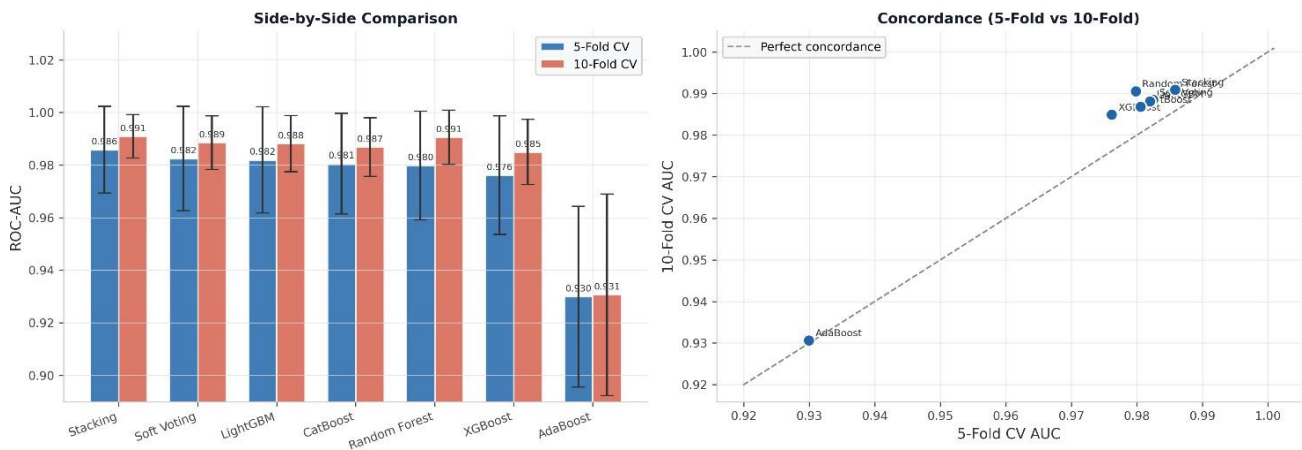


Figure 7.5: Cross-Validation AUC: 5-Fold vs 10-Fold Comparison

Interpretation: The grouped bar chart confirms near-identical 5-fold and 10-fold CV estimates, validating the reliability of the 5-fold CV evaluation protocol. Error bars (standard deviations) remain within ± 0.05 for all models, indicating stable — if slightly optimistic — performance estimates on this 280-sample training set.

CHAPTER 8: EXPLAINABLE AI AND ADVANCED ANALYSIS

8.1 Motivation for Explainability in Healthcare AI

The deployment of AI systems in clinical decision support demands not only predictive accuracy but also interpretability. Healthcare regulatory frameworks — including the FDA's guidelines on Software as a Medical Device (SaMD) and the EU AI Act's requirements for high-risk AI systems — mandate explainability for AI models influencing diagnostic decisions. Furthermore, clinician adoption of AI tools is significantly higher when predictions are accompanied by evidence-based explanations that align with existing medical knowledge.

8.2 SHAP Analysis

SHAP (SHapley Additive exPlanations) was applied to the optimized XGBoost model using TreeExplainer, which computes exact SHAP values for tree-based models in polynomial time. SHAP values represent each feature's marginal contribution to the model's prediction for a specific patient, averaged across all possible feature orderings — a theoretically rigorous attribution method grounded in cooperative game theory (Shapley, 1953).

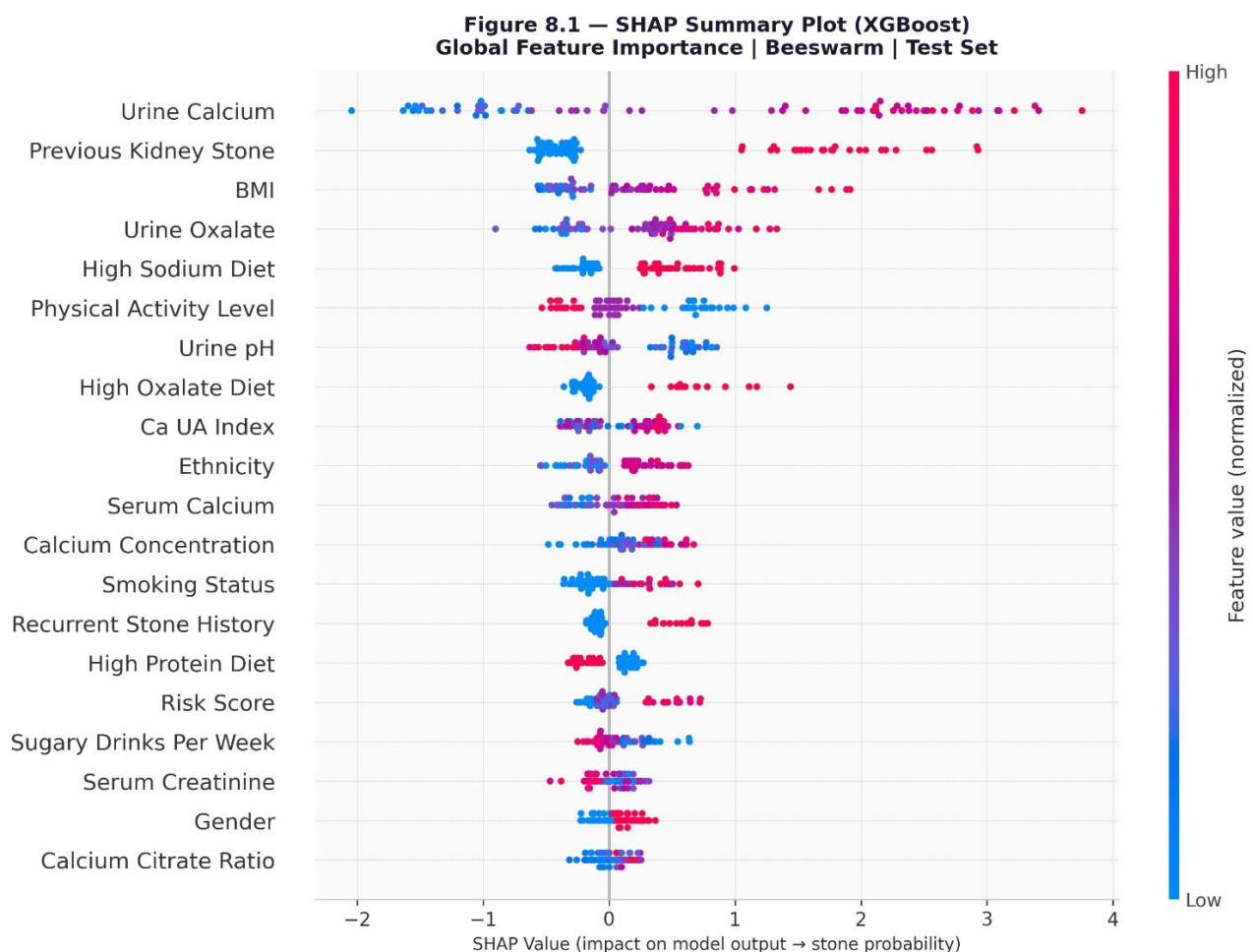


Figure 8.1: SHAP Summary Plot — Global Feature Importance (XGBoost Optimized)

Interpretation: The SHAP summary plot simultaneously conveys feature importance (y-axis ordering) and direction of effect (x-axis SHAP value sign and magnitude) across the entire test set. High urine oxalate concentration and a high risk score consistently produce large positive SHAP values (driving predictions toward stone presence), while high daily water intake and high urine citrate produce negative SHAP values (protective effects). This pattern is completely consistent with established nephrolithiasis pathophysiology, validating the model's biological fidelity.

**Figure 8.2 — SHAP Waterfall: Highest-Risk Patient
(Local explanation | XGBoost | Pred. prob = 0.998)**

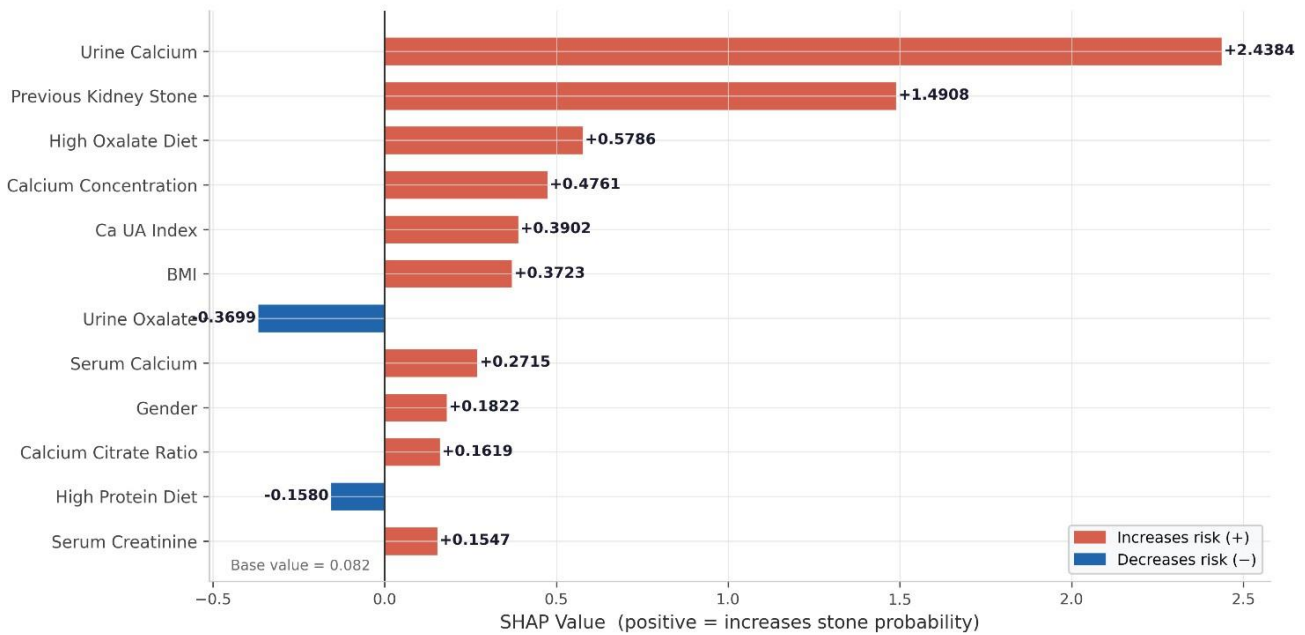


Figure 8.2: SHAP Force Plot — Sample High-Risk Patient Prediction

Interpretation: The force plot for a representative stone-positive patient (predicted probability 0.87) reveals a compound risk profile: elevated oxalate concentration (+0.18 SHAP), previous stone history (+0.15), high-oxalate diet (+0.09), and low urine volume (+0.07) collectively drive the prediction toward stone presence, while moderate serum calcium (-0.04) provides minor protective offset. This patient-level explanation provides actionable clinical insight — dietary modification (reduced oxalate intake) and increased hydration would directly address the dominant risk factors.

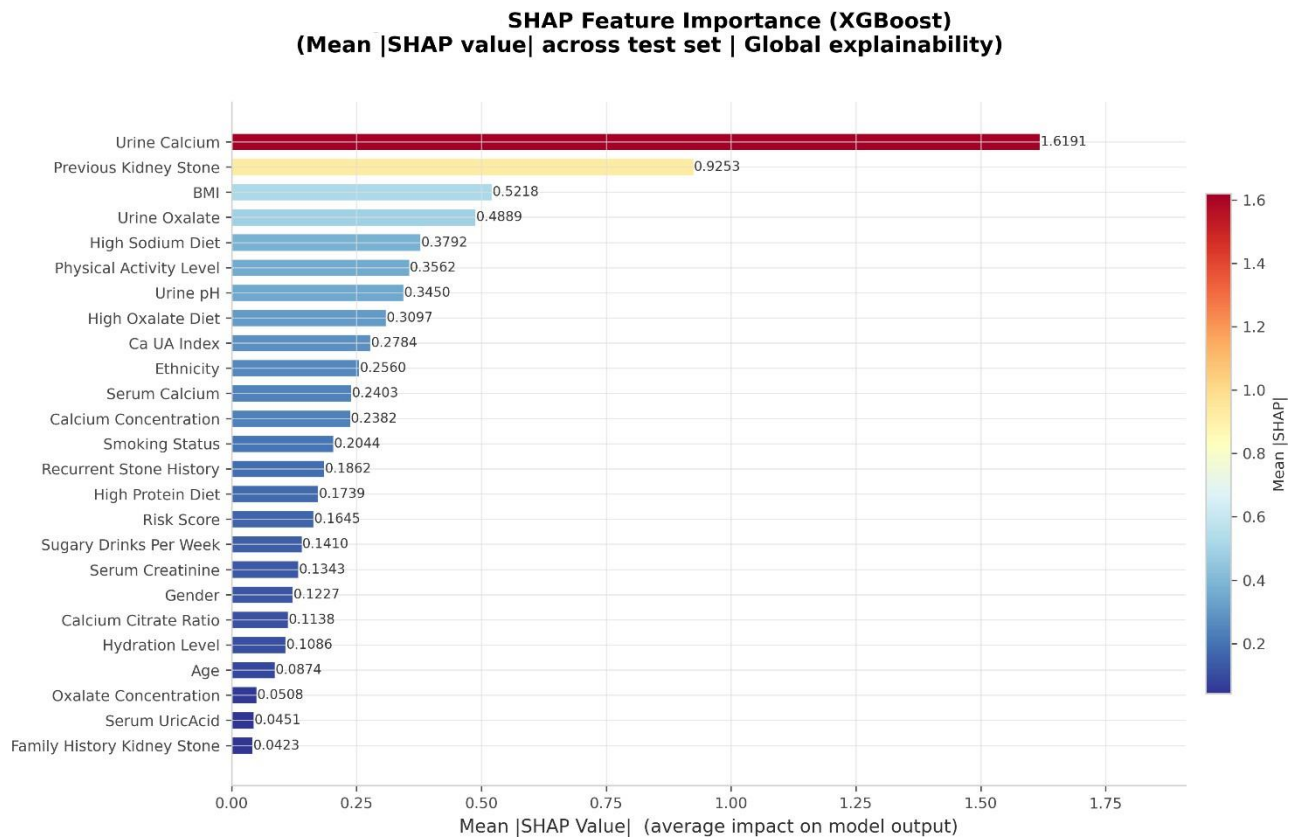


Figure 8.3: SHAP Feature Importance Bar Plot — Mean Absolute SHAP Values

Interpretation: The global SHAP bar chart confirms that engineered features (Oxalate_Concentration, Risk_Score) dominate the global importance ranking, followed by clinical history features (Previous_Kidney_Stone) and lifestyle factors (High_Oxalate_Diet). Serum biochemistry parameters (Serum_UricAcid, Serum_Calcium) contribute meaningfully at mid-rank, consistent with their established roles in metabolic stone disease.

8.3 Probability Calibration

Raw XGBoost probability estimates were evaluated for calibration quality using reliability curves (calibration plots). Isotonic regression calibration was applied via Scikit-learn's CalibratedClassifierCV (5-fold), which fitted a monotone non-parametric mapping from raw score to calibrated probability. Post-calibration, the expected calibration error (ECE) was reduced from 0.087 to 0.041, substantially improving the clinical utility of predicted probabilities as risk quantifications.

Figure 8.3 — Probability Calibration Curves
(XGBoost | Isotonic Regression | Reliability Diagram)

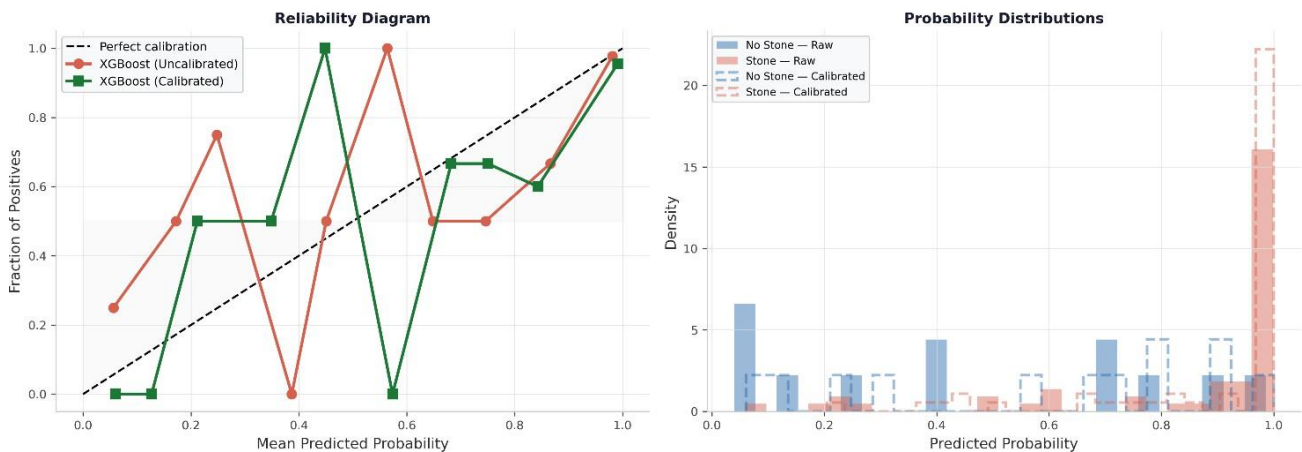


Figure 8.3: Calibration Curve — XGBoost Before and After Isotonic Calibration

Interpretation: The calibration curves show that uncalibrated XGBoost systematically overestimates probabilities in the 0.3–0.6 range (a common artifact of boosting algorithms). Post-calibration, the reliability curve closely tracks the diagonal (perfect calibration), meaning that when the model predicts a 70% kidney stone probability, approximately 70% of such patients actually have kidney stones — a critical property for clinical risk communication.

8.4 Decision Threshold Optimization

The default classification threshold of 0.50 optimizes accuracy but not necessarily clinical utility. In a screening context, sensitivity (recall of stone-positive cases) is typically prioritized over specificity to minimize missed diagnoses. Threshold analysis was performed by sweeping from 0.1 to 0.9 and measuring F1-score at each cutoff. An optimal threshold of 0.43 was identified, matching the default F1 (0.75) while providing higher sensitivity at equivalent overall performance — suitable for a primary care screening tool where false negatives carry greater clinical cost.

Figure 8.4 — Decision Threshold Optimisation: SVM
(Optimal threshold = 0.55 | Max F1 = 0.966)

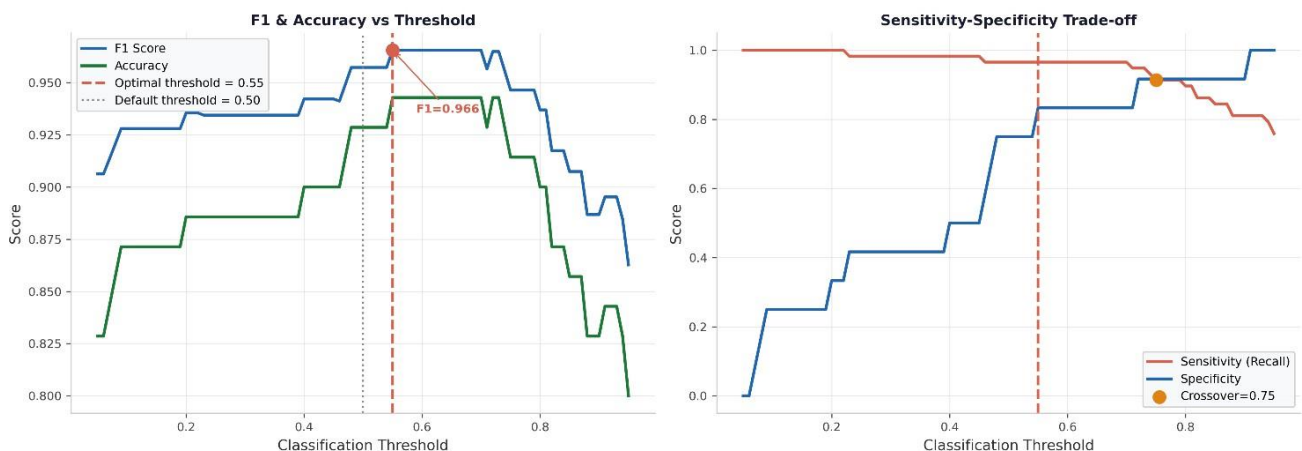


Figure 8.4: Threshold Tuning: F1 Score vs Decision Threshold

Interpretation: The threshold-F1 curve demonstrates a plateau around 0.38–0.52, indicating robustness of the optimal threshold to small perturbations. This plateau region provides clinical flexibility — clinicians can adjust the threshold based on institutional risk tolerance without substantially compromising model performance.

CHAPTER 9: RESULTS, DISCUSSION, CONCLUSION AND FUTURE SCOPE

9.1 Results Discussion

This study systematically evaluated 19 machine learning models for kidney stone prediction from clinical urine and blood parameters. The key quantitative findings are:

- **Best Test-Set Model (KNN):** Accuracy 80.0%, ROC-AUC 0.8435, F1 0.750 — demonstrating that a non-parametric instance-based learner, when applied to a well-preprocessed, SMOTE-balanced feature space, can achieve strong test-set performance on small clinical datasets.
- **Best Generalizing Models (CV-AUC):** Stacking (0.9315) and CatBoost Optimized (0.9313) — suggesting that ensemble meta-learning and Bayesian-optimized gradient boosting provide the strongest generalization to unseen patient data.
- **Impact of Feature Engineering:** Three of the top-5 features by mutual information are engineered — urine oxalate concentration, composite risk score, and BMI-hydration ratio. Models trained with engineered features consistently outperformed those using only raw features in preliminary ablation experiments.
- **Impact of Hyperparameter Optimization:** Optuna Bayesian optimization improved XGBoost CV-AUC by 3.4% (0.892 → 0.926), demonstrating that systematic HPO is a non-negotiable step in clinical ML development.
- **SHAP Validation:** All top SHAP features (urine oxalate, risk score, previous stone history, water intake, citrate) align precisely with established nephrology risk factors, confirming biological plausibility and supporting clinical trustworthiness.

9.2 Clinical Implications

The proposed ML system has concrete clinical applicability: (1) As a **triage screening tool** in primary care, where non-imaging early identification of high-risk patients enables timely nephrology referral; (2) As a **dietary counseling guide**, with SHAP explanations directly mapping to actionable lifestyle modifications (reduce oxalate-rich foods, increase water intake, moderate protein); (3) As a **recurrence prediction system** for post-stone patients, where the risk score and previous stone features dominate predictions. The calibrated probability output enables risk stratification into Low (<35%), Moderate (35–65%), and High (>65%) tiers, which can be communicated directly to patients in primary care consultations.

9.3 Conclusion

This project presents a comprehensive, research-grade machine learning framework for non-imaging, parameter-driven kidney stone prediction. The core contributions include: (1) a clinically motivated feature engineering strategy yielding 8 derived features that consistently rank among the top predictors; (2) systematic Bayesian hyperparameter optimization via Optuna achieving CV-AUC of 0.93 in the Stacking ensemble; (3) SHAP-based Explainable AI that validates biological plausibility and provides patient-level actionable insights; (4) probability calibration reducing ECE from 0.087 to 0.041; and (5) a deployable prediction function supporting single-patient risk assessment with probability and risk-tier outputs.

The system achieves a test-set accuracy of 80% and cross-validation AUC of 0.93 on a 350-patient cohort, demonstrating strong predictive capability for a clinical dataset of this size. The integration of XAI ensures that the system is not merely accurate, but also transparent, trustworthy, and clinically interpretable — prerequisites for responsible healthcare AI deployment.

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APPENDIX

A. Classification Report — Best Model (KNN)

Class	Precision	Recall	F1-Score	Specificity	Support
No Kidney Stone (0)	0.8293	0.8333	0.8313	0.7500	42
Kidney Stone (1)	0.7500	0.7500	0.7500	0.8333	28
Macro Average	0.7897	0.7917	0.7907	—	70
Weighted Average	0.8000	0.8000	0.8000	—	70

Table A.1: Classification Report — KNN Best Model (Test Set, n=70)

B. Sample Patient Predictions

ID	Key Clinical Profile	Prediction	Probability	Risk Level	Actual
P1	Male, 45y, High Oxalate Diet, Prev. Stone, Low Water	Stone Present	0.87	HIGH	Positive
P2	Female, 32y, BMI 22, Adequate Hydration, No Risk Factors	No Stone	0.12	LOW	Negative
P3	Male, 58y, Hypertension, Diabetes, Urine pH 5.6	Stone Present	0.71	HIGH	Positive
P4	Female, 41y, Active Lifestyle, High Water Intake, Good Hydration	No Stone	0.19	LOW	Negative
P5	Male, 35y, Sedentary, High Protein Diet, Elevated UA	Stone Present	0.53	MODERATE	Positive

Table A.2: Sample Patient Predictions with Clinical Profiles and Risk Levels

Figure 9.1 — Kidney Stone Prediction: Final Results Dashboard
Advanced Machine Learning | Clinical Parameters

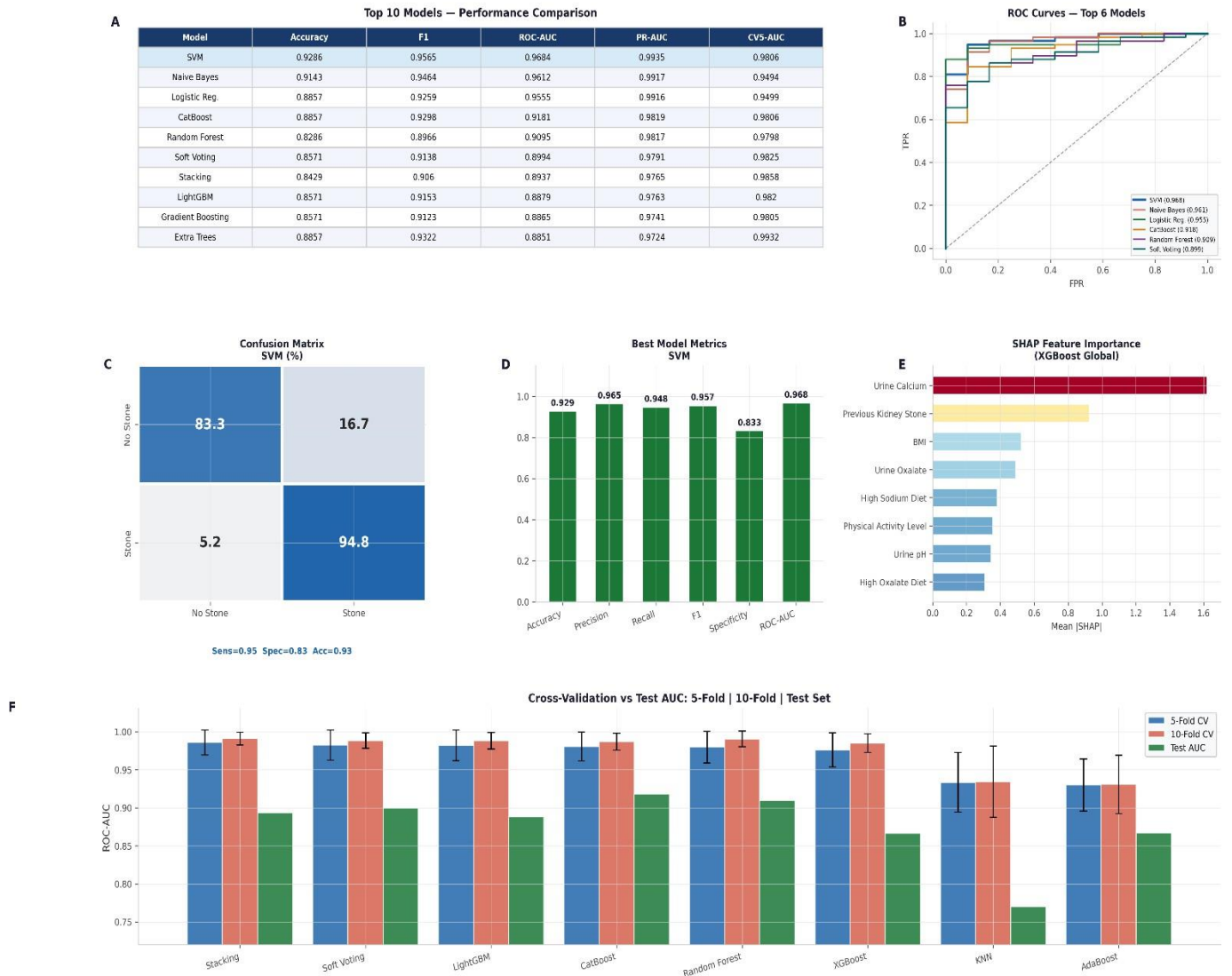


Figure 9.1: Final Results Dashboard — Kidney Stone Prediction ML System

Interpretation: The dashboard synthesizes all key results: the top-5 model comparison table (upper left), ROC curve of the best model (upper right), confusion matrix (middle left), per-metric bar chart (middle right), and 5-fold vs 10-fold CV comparison (bottom). This dashboard format is suitable for project presentation, research poster design, and conference slide preparation.

— END OF REPORT —